FACTORING

BASIC ALGEBRA

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

The Greatest Common Factor

Factoring by Grouping

Factoring Quadratics

Synthetic Division

THE GREATEST COMMON FACTOR

GREATEST COMMON FACTOR

The **largest** expression that divides each term

Factor $\mathbf{15 + 25x}$

GCF ,

$$\frac{15}{5} = 3 \qquad \frac{25x}{5} = 5x$$

$$\boxed{5\,(3 + 5x)}$$

ans

Check ,

$$5(3 + 5x) = (5 \cdot 3) + (5 \cdot 5x)$$

$$= 15 + 25x$$

Factor $x^2y + y^2x^3$

GCF ,

$$\frac{x^2y}{x^2y} = 1 \qquad \frac{y^2x^3}{x^2y} = xy$$

$$\boxed{x^2y\,(xy + 1)}$$

ans

Check ,

$$x^2y(xy + 1) = x^2y(xy) + x^2y(1)$$

$$= x^3y^2 + x^2y$$

FACTORING BY GROUPING

FACTORING BY GROUPING

Factor $x^3 + 3x^2 + 4x + 12$

$(x^3 + 3x^2) + (4x + 12)$

GCF ,

$\frac{(x^3 + 3x^2)}{x^2} = x + 3$ $\frac{(4x + 12)}{4} = x + 3$

Grouping terms (**4 terms**) with common factor

$x^2(x + 3) + 4(x + 3)$

FACTORING BY GROUPING

Grouping terms (**4 terms**) with common factor

Factor $x^3 + 3x^2 + 4x + 12$

$(x^3 + 3x^2) + (4x + 12)$

GCF ,

$\frac{(x^3 + 3x^2)}{x^2} = x + 3$ $\frac{(4x + 12)}{4} = x + 3$

$x^2(x + 3) + 4(x + 3)$

$\boxed{(x + 3)(x^2 + 4)}$

ans

Factor $2x^3 - 10x^2 + 7x - 35$

$$(2x^3 - 10x^2) + (7x - 35)$$

GCF ,

$$\frac{(2x^3 - 10x^2)}{2x^2} = x - 5 \qquad \frac{(7x - 35)}{7} = x - 5$$

$$2x^2(x - 5) + 7(x - 5)$$

Factor $2x^3 - 10x^2 + 7x - 35$

$$(2x^3 - 10x^2) + (7x - 35)$$

GCF ,

$$\frac{(2x^3 - 10x^2)}{2x^2} = x - 5 \qquad \frac{(7x - 35)}{7} = x - 5$$

$$2x^2(x - 5) + 7(x - 5)$$

$$\boxed{(x - 5)\ (2x^2 + 7)}$$

ans

FACTORING QUADRATICS

A quadratic expression is a **second-degree polynomial expression**

Factor $\boldsymbol{x^2 - 6x + 8}$

$(x \pm \quad)(x \pm \quad)$

$\boldsymbol{8}$

$\boldsymbol{-2}$ $\boldsymbol{-4}$

$\boldsymbol{-6}$

$\boldsymbol{1}$	$\boldsymbol{8}$
$\boldsymbol{2}$	$\boldsymbol{4}$
$\boldsymbol{-2}$	$\boldsymbol{-4}$

$\boxed{(x-2)(x-4)}$

ans

FACTORING QUADRATICS

A quadratic expression is a **second**-degree polynomial expression

Factor $\boldsymbol{x^2 - 6x + 8}$

Check ,

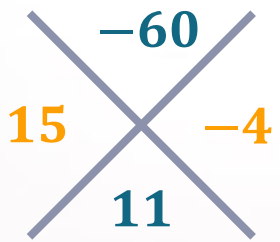
$$(x-2)(x-4)$$

$$(x \cdot x) + (x \cdot -4) + (-2 \cdot x) + (-2 \cdot -4)$$

$$x^2 \ -4x \ -2x \ \ + 8$$

$$x^2 - 6x + 8$$

Factor $10x^2 + 11x - 6$



−1	60
−2	30
−3	20
−4	15

$10x^2 - 4x + 15x - 6$

Factoring by grouping

$(10x^2 - 4x) + (15x - 6)$

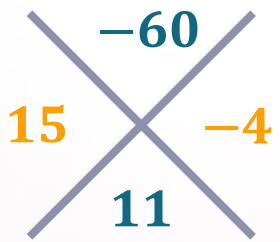
GFC ,

$$\frac{(10x^2 - 4x)}{2x} = 5x - 2$$

$$\frac{(15x - 6)}{3} = 5x - 2$$

$2x(5x - 2) + 3(5x - 2)$

Factor $10x^2 + 11x - 6$



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$10x^2 - 4x + 15x - 6$

Factoring by grouping

$(10x^2 - 4x) + (15x - 6)$

GFC ,

$$\frac{(10x^2 - 4x)}{2x} = 5x - 2$$

$$\frac{(15x - 6)}{3} = 5x - 2$$

$2x(5x - 2) + 3(5x - 2)$

$$\boxed{(5x - 2)\,(2x + 3)}$$

ans

DIFFERENCE OF SQUARES

$$a^2 - b^2 = (a + b)(a - b)$$

Factor $x^2 - 16$

$x^2 - 4^2$

$a^2 - b^2$

$\boxed{(x + 4)\,(x - 4)}$

ans

Factor $9p^2 - 1$

$(3p)^2 - 1^2$

$a^2 - b^2 = (a + b)(a - b)$

$\boxed{(3p + 1)\,(3p - 1)}$

ans

Sum of squares **DOES NOT** factor

$x^2 + 2^2 \neq (x + 2)(x - 2)$

DIFFERENCE OR SUM OF CUBES

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

Factor $y^3 - 27$

$$y^3 - 3^3 = (y - 3)\,(y^2 + y3 + 3^2)$$

$$= \boxed{(y - 3)(y^2 + 3y + 9)}$$

ans

Factor $y^3 + 27$

$$y^3 + 3^3 = (y + 3)\,(y^2 - y3 + 3^2)$$

$$= \boxed{(y + 3)(y^2 - 3y + 9)}$$

ans

Which of these **DOES NOT** factor?

A. $x^2 + x$

B. $x^2 - 25$

C. $x^2 + 4$

D. $x^3 + 2x^2 + 3x + 6$

E. $5x^2 - 14x + 8$

Which of these **DOES NOT** factor?

A. $x^2 + x$ → $\boxed{x(x+1)}$

B. $x^2 - 25$

C. $x^2 + 4$

D. $x^3 + 2x^2 + 3x + 6$

E. $5x^2 - 14x + 8$

Which of these **DOES NOT** factor?

A. $x^2 + x$

B. $x^2 - 25$ $\longrightarrow$ $\boxed{(x+5)(x-5)}$

C. $x^2 + 4$

D. $x^3 + 2x^2 + 3x + 6$

E. $5x^2 - 14x + 8$

Which of these **DOES NOT** factor?

A. $x^2 + x$

B. $x^2 - 25$

C. $\boxed{x^2 + 4}$ → *DOES NOT factor*

ans

D. $x^3 + 2x^2 + 3x + 6$

E. $5x^2 - 14x + 8$

Which of these **DOES NOT** factor?

A. $x^2 + x$

B. $x^2 - 25$

C. $x^2 + 4$

D. $x^3 + 2x^2 + 3x + 6$ →

E. $5x^2 - 14x + 8$

Factoring by grouping

$$(x^3 + 2x^2) + (3x + 6)$$

GFC :

$$\frac{(x^3 + 2x^2)}{x^2} = x + 2 \qquad \frac{(3x + 6)}{3} = x + 2$$

$$x^2(x + 2) + 3(x + 2)$$

Which of these **DOES NOT** factor?

A. $x^2 + x$

B. $x^2 - 25$

C. $x^2 + 4$

D. $x^3 + 2x^2 + 3x + 6$ →

E. $5x^2 - 14x + 8$

Factoring by grouping

$$(x^3 + 2x^2) + (3x + 6)$$

GFC :

$$\frac{(x^3 + 2x^2)}{x^2} = x + 2 \qquad \frac{(3x + 6)}{3} = x + 2$$

$$x^2(x + 2) + 3(x + 2)$$

$$\boxed{(x + 2)(x^2 + 3)}$$

Which of these **DOES NOT** factor?

A. $x^2 + x$

B. $x^2 - 25$

C. $x^2 + 4$

D. $x^3 + 2x^2 + 3x + 6$

E. $5x^2 - 14x + 8$ $\longrightarrow$

Factoring by grouping

$$\begin{array}{ccc} & 40 & \\ -10 & & -4 \\ & -14 & \end{array}$$

$$5x^2 - 10x - 4x + 8$$

$$(5x^2 - 10x) - (4x - 8)$$

Which of these **DOES NOT** factor?

A. $x^2 + x$

B. $x^2 - 25$

C. $x^2 + 4$

D. $x^3 + 2x^2 + 3x + 6$

E. $5x^2 - 14x + 8$ $\longrightarrow$

Factoring by grouping

$$(5x^2 - 10x) - (4x - 8)$$

GCF :

$$\frac{(5x^2 - 10x)}{5x} = x - 2 \qquad \frac{(4x - 8)}{4} = x - 2$$

$$5x(x - 2) + 4(x - 2)$$

Which of these **DOES NOT** factor?

A. $x^2 + x$

B. $x^2 - 25$

C. $x^2 + 4$

D. $x^3 + 2x^2 + 3x + 6$

E. $5x^2 - 14x + 8$ ⟶

Factoring by grouping

$$(5x^2 - 10x) - (4x - 8)$$

GCF :

$$\frac{(5x^2 - 10x)}{5x} = x - 2 \qquad \frac{(4x - 8)}{4} = x - 2$$

$$5x(x - 2) - 4(x - 2)$$

$$\boxed{(x - 2)\,(5x - 4)}$$

SYNTHETIC DIVISION

A **shorthand method** of dividing polynomials for the special case of dividing by a **linear factor** whose leading coefficient is 1

Factor $x^3 + 4x^2 - 11x - 30$

$\cancel{1}$	1	4	−11	−30
		1	5	−6
	1	5	−6	−36

$\cancel{2}$	1	4	−11	−30
		2	12	2
	1	6	1	−28

A **shorthand method** of dividing polynomials for the special case of dividing by a **linear factor** whose leading coefficient is 1

Factor $x^3 + 4x^2 - 11x - 30 = 0$

$-2\|$	1	4	-11	-30
		-2	-4	30
	1	2	-15	0
	x^2	$2x$	-15	

$x = -2$

$x + 2 = 0$

$(x + 2)\left(x^2 + 2x - 15\right)$

$(x + 5)(x - 3)$

A **shorthand method** of dividing polynomials for the special case of dividing by a **linear factor** whose leading coefficient is 1

Factor $x^2 + 2x - 15$

3|	1	2	-15
		3	15
	1	5	0
	x	5	

$x = 3$

$x - 3 = 0$

$$\boxed{(x+2)\,(x-3)\,(x+5)}$$

ans

SEATWORK